

An audit of the 2026 Navier–Stokes finite-time blowup claim:
the formal statement read against Clay’s wording and Mathlib’s
definitions,
the Lean proof built and its axioms asked of the kernel, and what
remains open

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machine-derived, not peer-reviewed; author list to be agreed

Abstract

On 8 September 2026 OpenAI published a 166-page proof, with a Lean 4 formalisation, that for every positive viscosity a smooth compactly supported force drives the three-dimensional incompressible Navier–Stokes flow from rest to unbounded velocity in finite time, and stated that this establishes alternatives (C) and (D) of the Clay problem statement [4, 2, 1]. This note records an audit made the following day from a corpus pinned by digest: the formal statement read hypothesis by hypothesis against Fefferman’s wording and Mathlib’s definitions; the 618,762-line repository built on a laptop and the kernel asked for the axioms of each main declaration; Comparator run with a second kernel; the paper mapped onto the certificate clause by clause; the writeup dissected by six independent readings and set against every classical obstruction; every printed identity a computer can decide re-run with red controls. What holds: the statement is Clay’s with nothing added or dropped, the build passes on the three standard axioms, both kernels accept, and no error or counterexample was found. Not decided: the paper’s finite-energy clause has no formal counterpart; the smallness parameter is bounded numerically nowhere; 3 checks of the companion Euler paper are listed and unrun. The register row reads PARTIAL. Nothing is claimed about the mathematics beyond what the kernel and the readings establish, nor about priority.

1 Introduction

The claim is Theorem 1.1 of [2]: for every $\nu > 0$ there are a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$, a compact $K \subset \mathbb{R}^3$ and smooth (u, p) on $\mathbb{R}^3 \times [0, 1)$, supported in K , solving the Navier–Stokes equations from rest, with $\sup_{0 \leq t < 1} \|u(t)\|_{L^2} < \infty$ and $\limsup_{t \uparrow 1} \|u(t)\|_{L^\infty} = \infty$; consequently no smooth solution on $\mathbb{R}^3 \times [0, \infty)$ with the same force and datum has uniformly bounded kinetic

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energy. The paper says this is alternative (C) of the Clay problem statement [1] and, through its Corollary 10.6, alternative (D). A Lean 4 repository [5] formalises both; the announcement [4] says the proof was produced by an internal model and formalised by another.

A formal certificate changes what an audit of such a claim can be: three questions become decidable by machine and one by reading. Is the formal statement the theorem the prose names; does the proof term check, and on which axioms; does the paper say what the certificate says; and where the paper asserts rather than displays, does the certificate discharge the assertion? The position throughout is the one the records state: *the Lean certificate is the claim; the paper is its human-readable evidence*. A gap in the paper is a gap in the writeup unless the Lean shows the same, and a printed identity that failed would refute the formula, not the theorem [19]. The contribution is not a theorem but the grammar of the audit and its public record; every number below is interpolated from a record by a tool that refuses when a record no longer supports a sentence. Sections 2–6 give the claim and the corpus, the audit, what the record decides, what is not claimed, and the dates as the parties state them, weighed by nobody here.

2 What was claimed, and what was pinned

Clay’s alternatives. Fefferman’s statement [1] asks for a proof of one of four assertions; in its wording, (C) *Breakdown of Navier–Stokes solutions on $\mathbb{R}^3$* : take $\nu > 0$ and $n = 3$; then there exist a smooth, divergence-free vector field $u^\circ(x)$ on $\mathbb{R}^3$ and a smooth $f(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$, satisfying (4), (5), for which there exist no solutions (p, u) of (1), (2), (3), (6), (7) on $\mathbb{R}^3 \times [0, \infty)$; and (D) the same on $\mathbb{R}^3/\mathbb{Z}^3$ with (8), (9) in place of (4), (5) and (10), (11) in place of (6), (7), the errata adding that the pressure is periodic. The numbered conditions are those of Table 1.

The mechanism. As the records describe it [19]: a self-similar axisymmetric core with radial scale $\tau^{1/2}$, axial scale $\tau^{1/2-h}$ and azimuthal and axial speeds $\tau^{-1/2-h}$, $\tau = 1 - t$, $0 < h < 1/100$, whose momentum residual is unbounded in an annulus; two families of shear-amplified oscillatory pulses on an auxiliary torus that supply the missing stress through their averaged quadratic flux; corrections at every order in q^{2h} ; an exact radial-heat exterior; cutoffs. The force is *defined* as the residual of the constructed flow and shown to extend smoothly through $t = 1$; uniqueness among smooth finite-energy solutions with a merely smooth pressure transfers the growth to any competitor, in the forced-blowup program of Córdoba and Martínez-Zoroa [9].

The corpus. Everything read was pinned on 2026-09-09 with its digest: 24 files: the two papers, the Clay statement, the Lean bridge files, the Formal Conjectures statement, the announcement and the concurrent papers and statements [2, 3, 4, 5, 6, 10]. The repository [5] has one commit, 8937a8f4, at 2026-09-08 10:57 UTC; it pins Lean v4.34.0-rc2 [7] and Mathlib at 85e3a25e [7]; it holds 2,486 files and 618,762 lines, of which 404,869 lines in 644 files are the Navier–Stokes library, 213,419 in 1,840 the Euler library and 474 in 2 the challenge statements. Its `formalization.yaml` declares the review *self-assessed* and the automation *GPT-6 Astra*, with no sorry [19].

3 The audit

3.1 The statement, hypothesis by hypothesis

The theorem the certificate proves was not written by the prover. The Comparator challenge module is the Formal Conjectures statement of the Clay alternatives [6], added by DeepMind on 2026-05-15 (pull request #1457), last touched on 2026-07-27 at revision 8bf45ed7, the revision OpenAI pinned and the revision on `main` on the day of the audit. OpenAI’s copy differs from upstream in 4 lines, every one an `open` or a notation, and the solution side re-declares the same definitions under the same names, byte-identical once comments are stripped [19]. Comparator compares every constant reachable from the theorem’s type for exact equality, so the only question is whether that statement is Clay’s.

Both theorems have the shape $\exists u_0 f, H(u_0) \wedge H(f) \wedge \neg \exists v p, S(v, p)$: the prover must *establish* H for the constructed data and *exclude* every (v, p) in S . A junk value or a weak definition inside S only enlarges the class and strengthens the theorem; a gap could hide only in an S stronger than Clay’s (6)(7) or (10)(11), excluding competitors Clay admits, or in an H weaker than Clay’s (4)(5) or (8)(9), admitting data Clay rejects. Every row of Table 1 is read in that direction, with the Mathlib definition located at the pinned revision [19].

Proposition 1 (statement fidelity; the record’s verdict). *The formal statements of (C) and (D), `navier_stokes_breakdown_R3` and `navier_stokes_breakdown_periodic`, are Clay’s with no hypothesis added to the competitor class and none dropped from the data; in particular no decay, growth or integrability condition is imposed on the pressure or on ∇v beyond smoothness, so the formal uniqueness argument must handle every competitor Clay admits. The dimension is `Fin 3` and ν is any positive real.*

A negation is cheap if the class it negates is empty for a trivial reason, so a witness was written for this audit and compiled against OpenAI’s challenge module: the fluid at rest satisfies every field of both competitor structures and the zero data all four data conditions (`cert-machine-NonVacuity.lean`, 7 declarations, 0 errors, the three standard axioms for `classes_inhabited`) [19]. It shows the definitions are satisfiable, not that they admit every classical solution, which is Table 1’s job.

3.2 The build, the axioms and the second kernel

The repository was built on 2026-09-09 on an ordinary laptop (Mac14,2, 8 cores, 16 GB, darwin 25.4.0, arm64) with the pinned toolchain: all 2,486 modules in 140 minutes after the Mathlib cache (11,251 Lake jobs in all; two legs, 681 modules at 8 threads then 1,805 at 4, after eight parallel Lean processes drove the machine into swap), exit 0, no errors [19]. A scratch file importing the two solution modules then asked `#print axioms` of the 4 declarations `formalization.yaml` names, and the kernel reported exactly `propext`, `Classical.choice`, `Quot.sound` for each. In code lines, comments blanked, the proof libraries contain no `sorry` beyond the 4 intentional placeholders of the challenge files, no axiom declaration, no `native_decide`, `unsafe`, `opaque` or `partial def`, no custom syntax and no heartbeat override (5 docstrings record earlier overrides removed).

Then the judge. Comparator [8], the Lean FRO’s tool for exactly this situation, exports both sides with `lean4export`, compares every constant reachable from each theorem’s type for exact equality, walks the axioms, and replays the proof through Lean’s kernel and through

Clay	Lean (Formal Conjectures) and the Mathlib definition at 85e3a25e	direction
(1) $\partial_t u + (u \cdot \nabla)u = \nu \Delta u - \nabla p + f$ on $\mathbb{R}^3 \times [0, \infty)$	<code>derivWithin (v x ·) (Ici 0) t + fderiv ℝ (v · t) x (v x t) = nu • Δ (v · t) x - gradient (p · t) x + f</code> <code>x t, ∀x, ∀t ≥ 0</code> : the one-sided time derivative at $t = 0$ within $[0, \infty)$; <code>fderiv</code> applied to v is $(v \cdot \nabla)v$; Δ the ordinary Laplacian (trace of the second derivative); <code>gradient</code> the Riesz representative of <code>fderiv</code>	same
(2) $\operatorname{div} u = 0$	$\nabla \cdot (v \cdot t) x = 0$, the trace of the Jacobian	same
(3) $u(x, 0) = u^\circ$	<code>v x 0 = u_0 x</code>	same
(6) $p, u \in C^\infty(\mathbb{R}^3 \times [0, \infty))$	<code>ContDiffOn ℝ ∞ (v) (univ ×^s Ici 0)</code> , likewise $p: C^\infty$	same
(7) $\int u ^2 dx < C$ for all t	up to $t = 0$ in the Taylor-series sense, not analytic <code>integrable : ∀ t ≥ 0, MemLp (v · t) 2</code> with <code>globally_bounded_energy : ∃ E, ∀ t ≥ 0, ∫ v x t ² < E</code> ; the Bochner integral of a non-integrable function is 0, which the first field forbids	same
(10) u periodic; errata: p periodic; (11) smooth	<code>IsOnePeriodic (v · t)</code> , <code>IsOnePeriodic (p · t)</code> for $t \geq 0$, along <code>EuclideanSpace.single i 1</code> ; smoothness as (6)	same as Clay plus the errata
(4) u° smooth, divergence-free, $ \partial^\alpha u^\circ \leq C(1 + x)^{-K}$ $\forall \alpha, K$	<code>InitialVelocityConditionDecay</code> : divergence-free, <code>ContDiff ℝ ∞ u_0, ∀m K ∃C, iteratedFderiv ℝ m u_0 x ≤ C/(1+ x)^K</code> ; the operator norm of the m -th derivative is comparable to the largest order- m partial	equivalent; the paper takes $u^\circ = 0$
(5) f smooth on $\mathbb{R}^3 \times [0, \infty)$, $ \partial_x^\alpha \partial_t^m f \leq C(1 + x + t)^{-K}$	<code>ForceConditionDecay: ContDiffOn ℝ ∞ (f) (univ ×^s Ici 0)</code> and the same bound on <code>iteratedFderivWithin</code> of total space-time order m in place of Clay’s split (α, m)	equivalent; a C_c^∞ force satisfies it trivially
(8) u°, f periodic in x ;	<code>InitialVelocityConditionPeriodic</code> ;	same
(9) $ \partial^\alpha \partial_t^m f \leq C(1 + t)^{-K}$	<code>ForceConditionPeriodic</code> with the time-decay bound on <code>iteratedFderivWithin</code>	

Table 1: Clay’s hypotheses against the formal statement, one row per hypothesis (9 rows; 7 read *same*, 2 *equivalent*). The first six rows are the competitor class S , the last three the data hypotheses H . Transcribed from the record [19].

nanoda, a kernel written independently of Lean’s in Rust and built from source for this run. On the 2 configurations, Navier–Stokes and Euler, all 4 kernel replays accept and 2 of 2 return “*Your solution is okay!*” with exit 0, in 13 and 17 minutes. One component did not run and the record says so: Comparator’s Landlock process isolation is Linux-only and a pass-through stand-in took its place; the statement comparison, the axiom walk and both kernel replays were performed, the isolation was not, and on a machine where both sides come from one pinned commit it guards against nothing this audit needed. What the build establishes is bounded by its trusted base, the two kernels, Mathlib’s definitions at the pinned revision, the elaborator and the hardware: the proof terms exist and check against the third party’s statement, and nothing about the paper.

3.3 The paper against the certificate

Each clause of the stated theorem was located in the formal library or named as absent (Table 2; 7 rows in the record, 2 of them not formal) [19].

One difference bears on the announcement, which says the fluid’s “energy remains finite through the entire dynamics, from rest to the formation of the singularity” [4]. Theorem 1.1

item	the paper	the Lean	status
the two theorems	Theorem 1.1 (C), Corollary 10.6 (D)	<code>navier_stokes_breakdown_R3</code> , <code>_periodic</code> , through <code>ComparatorBridge</code> adapters; solution-side definitions byte-identical to the challenge	proved, subject to the build record
uniqueness, Lemma 10.5	v smooth, $L_t^\infty L_x^2$, zero datum, same force $\Rightarrow v = u$; no growth condition on the pressure	<code>classical_uniqueness_on_Icc</code> (<code>R3/WholeSpaceUniqueness.lean:30</code>): competitor smooth, divergence-free, uniformly finite energy, pressure only smooth; pressure recovered (<code>R3/PressureRecovery.lean:419</code>), flux bounded uniformly in the radius (<code>R3/PressureFlux.lean:576</code>)	covers every competitor Clay admits
the viscosity rescaling	(10.22): space rescaled, singular time 1	time rescaled, $f_\nu(x, t) = \nu^2 f(\nu t, x)$ (<code>ComparatorR3Theorem.lean:19</code>), singular time $1/\nu$	equivalent; the statement names no time
the uniform energy bound	Theorem 1.1's $\sup_{t < 1} \ u(t)\ _{L^2} < \infty$; Lemma 10.4; the announcement's “its energy remains finite through the entire dynamics”	<code>uniform_finite_energy</code> (<code>R3/CompactEnergy.lean:343</code>) proves it for any smooth compactly supported force and is used by nothing; the structure carrying <code>energy_bounded</code> has no producer; the candidate's <code>Properties</code> structure has ten fields and no energy field	NOT FORMAL; every other clause of Theorem 1.1 is
the force's support in time	$f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty))$: zero near $t = 0$	smooth for $t \geq 0$, compact spatial support, zero for $t \geq T$; no field states $f = 0$ near $t = 0$	weaker in Lean; the challenge's force condition does not require it
the growth	(10.20)–(10.21): $u_\theta = \tau^{-A}(e_0 + O(\tau^{2h}))$ along a named off-axis path	<code>SpeedUnboundedAtOne</code> : $\forall M \delta \exists t x$, $1 - \delta < t < 1$, $M < \ u(t, x)\ $; on the torus also $\ u(t, 0)\ \rightarrow \infty$ on the axis	pointwise $\limsup \ u\ _\infty =$ ∞ ; no rate
the numbering	Propositions 5.5, 7.5, 9.6, 9.9, Theorem 3.1	docstrings cite a manuscript numbering absent from the published paper	the map is by content

Table 2: The paper mapped onto the certificate, from the record [19].

states it and Lemma 10.4 proves it in a few lines, re-derived in the reading; the repository contains the general lemma, nothing invokes it, and `breakdownStatement`, the file's own “full assertion of Theorem 1.1”, is never proved. Clay's (C) does not need the clause, since a global smooth competitor is excluded by uniqueness on every $[0, T]$, $T < 1$, where the candidate's energy is finite by compact support alone; so the formal result stands as Clay's and the physically striking clause rests on the paper. The Euler theorem in the same repository carries its energy bound (Section 3.8); the record reads the asymmetry as an oversight rather than an obstacle, and this note repeats that reading as a reading.

3.4 The writeup, read

The paper was taken apart the day after it appeared by six independent readings, five section memos (§3 with §10, §4 with Appendix B, §5 with Appendix A, §6 with §7, §8 and §9 with Appendix C) and one map of the Lean library against all of them, each from the pinned PDF, written in the machine-assisted setting the author's disclosure describes and consolidated into one

record [19]. Each reading transcribed the statements, traced the order in which every parameter is fixed and re-derived what it could; no circularity and no error were found. What was verified by hand holds as printed in all 6 groups of the record: the closing argument of §10 term by term; the leading equations of §4 re-derived in the similarity coordinates; the heat exterior of Appendix A with Lemma A.8’s two zero-integral identities; the identities of §§6–7; the moment system of §8 and the loop of Lemma C.1 on 371 random relaxed-cone data, where choosing δ_L from the base point alone breaks the cone in about 7% of trials, so the ordering “ δ_L after $\mu_{\max}$, uniformly” is load-bearing; and the coefficient equations of §5, by computer (Section 3.7).

What the paper asserts rather than displays was listed and ranked by how much rests on it, 13 points in all [19]. Point 0 is the subject of Section 3.6; the other 12, in the record’s order, are the endpoint regularity of the local field up to $\tau = 0$ in Proposition 9.9 (one paragraph, no estimate displayed, on which the force’s smoothness at $t = 1$ rests); the uniformity in η of the flatness (3.4) that (10.9) needs; the admissibility of next-cycle sources in Proposition 9.3(i), the one unstated induction hypothesis of §9; the smallness parameters, every one “sufficiently small” and none numeric, the printed $h < 1/100$ not being the operative bound (Lemma 4.8 needs $h < \min\{\lambda, e^{-T_d}\}$ with $T_d = e^{M_d} + 10$, the derivative scale of §6 exceeds one only from dyadic level about 3×10^8 , and flatness to order N needs more than $1000(N + K_m)$ cycles and about $10^7 N$ integrations by parts); the Lipschitz bounds of Proposition B.2; the inequality $2 + h < a$ of Lemma 4.9 on the terminal collar; the continuation of Proposition B.5; the analyticity region of Lemma 5.1 against Theorem 4.6(i); the coefficient equations (5.3)–(5.6), since discharged by computer algebra; the two correction routes of §7 and §8.5; the hypothesis $|T_0| \geq c\zeta$ of Proposition C.3, never cited in §9; and the overloading of q_* . None is an error. 5 bookkeeping slips were found, each harmless with a positive margin (a dropped $-\kappa_s$ in the zero harmonic of two-wave products has margins 0.08 and 0.4 against $\kappa_s = 10^{-5}$). One is a warning to anyone auditing from extracted text: the cutoff family of Lemma 10.5 on p. 122 extracts as the dilations $\varphi_{4R}, \varphi_{8R}$, under which three inequalities are false, while the glyphs are the powers φ_R^4, φ_R^8 , under which every step is exact; no inequality is a finding until its glyphs have been seen.

3.5 The asserted steps, traced into the kernel

Each asserted step was traced into the Lean library at the pinned commit and asked a harsher question than “is it discharged”: whether the formal statement there is *weaker* than the paper’s claim [19]. 6 are discharged by a proved theorem at full strength, 2 of them stronger than the paper: the endpoint regularity is a genuine C^∞ extension across $t = 1$ (`OneSidedExtension`, `JointResidualLimits.lean:72`) and the flatness a joint space-time limit, η -uniform by construction (`VanishingJointJets`, `JointResidualLimits.lean:84`), exactly the uniformity the reading found missing. 5 are discharged with something altered, among them the smallness parameters: h is not a number in Lean but `Classical.choice` of a nonempty profile datum, constrained by $2h < \lambda < 1/10$ and the hypothesis $h \leq 1/1000$ (`NaturalAxisData.lean:41`); N , Λ , C , q_* and the cycle schedule are existentials; the only numeric constants are $\kappa = 10^{-5}$, $\Lambda_{\text{chart}} = 4 - \sqrt{2}$, $X_{\text{big}} = 100$ and $X_i = 110$; the Lipschitz machinery of Proposition B.2 and the analyticity region are proved as machinery, their instantiation for the actual profile not traced to a line. 1 point, Lemma 4.9’s collar inequality, has no counterpart, the formal tail being a different plateau construction; 3 of the slips are prose only, and two of the paper’s arithmetic slips are absent from the formal object.

Two more answers were read from the source. The construction’s non-axisymmetry, which Section 3.6 needs, is a theorem, `angularMode_ne_zero` (`ActualInitialization.lean:52`), not

an assumption. And the growth witness, `FinalSlowBase.axis_tendsto`, is the *axial* component on the axis along $t \mapsto (t, 0)$, not the paper’s off-axis azimuthal path (10.20)–(10.21); both give $\limsup \|u\|_\infty = \infty$, and no Lean theorem bounds the swirl $r u_\theta$ from below.

3.6 The adversarial pass

The question an audit of a singularity must ask is whether a theorem says this cannot happen. 9 classical obstructions were set against the construction’s own exponents [19]. Of these, 6 are necessary conditions and are met: Serrin’s $\int \|u\|_\infty^2 dt = \infty$ [13], with $\|u\|_\infty^2 \asymp \tau^{-1-2h}$; Escauriaza–Seregin–Šverák’s $\|u(t)\|_{L^3} \rightarrow \infty$ [16], met slowly, as $\tau^{-4h/3}$; the Beale–Kato–Majda integral [15]; Leray’s lower bounds [12]; finite energy and finite dissipation, the dissipation rate $\tau^{-1/2-3h}$ being integrable exactly when $h < 1/6$; and the Caffarelli–Kohn–Nirenberg measure bound [14], one point at one time. The other 3 are exclusions and are evaded: the type-I exclusions for axisymmetric flows [18, 17] twice, the flow being type II by the factor τ^{-h} and not axisymmetric, and the classical no-swirl regularity, the force injecting swirl.

The third is load-bearing and unstated. For an axisymmetric flow the swirl $\Gamma = r u_\theta$ satisfies

$$\partial_t \Gamma + u_r \partial_r \Gamma + u_z \partial_z \Gamma = \Gamma_{rr} - \Gamma_r/r + \Gamma_{zz} + r f_\theta,$$

a drift–diffusion equation with no zeroth-order term, derived in the battery from the θ -momentum equation and checked symbolically; so from rest with a bounded compactly supported force, $\sup |\Gamma(t)| \leq \int_0^t \sup |r f_\theta|$ and $|u_\theta| \leq C/r$ for every $t < 1$. The paper’s leading field has $r u_\theta^{(0)} = q^{-h} H$, in its own words on p. 27, unbounded as $q \downarrow 0$. The axisymmetric background alone is therefore impossible as a forced flow from rest; the theorem survives because the physical field is not axisymmetric, the pulses carrying nonzero integer angular frequencies (p. 12) whose Reynolds flux breaks the maximum principle for the angular mean. The paper does not say this anywhere; it is the sentence a referee would ask for, and it is why $h > 0$ is forced rather than chosen. The formal object evades the obstruction by `angularMode_ne_zero` (Section 3.5). This is a reading, not a refutation: the record is explicit that no obstruction refutes the construction.

3.7 The printed identities, re-run

The proof carries no numerics and fixes no constant to a number, so there is nothing for an interval certifier to certify. What the paper prints are identities, and each was re-run from the formula as printed by 7 probe scripts carrying 26 red controls, a deliberately wrong variant of each test that must be rejected, 0 of which failed to fire [19]. Symbolically: the similarity-coordinate derivatives of Lemma 4.1, the commutator kernel norm $16\pi/R$ of Lemma 10.5, the derivative count (10.12), the rescalings (10.22)–(10.23) and of Corollary 10.6, the exterior q -cancellation, the cutoff bound (10.3), and the exact exterior swirl $K = (r^2/2)^{-A} H(4\tau/r^2)$ against its equation (A.37). At 25 digits: the integral H of (A.32) against (A.37), worst relative residual 4.7×10^{-19} , and against its Taylor coefficients (A.35) to order 4. By exact algebra: the expansion (5.1) substituted into the axisymmetric operator and compared term by term with the printed coefficient equations (5.2)–(5.6), the sparsity and block nilpotency of A_1 in (5.7) and the divisibility of Ω_k by X , 69 identities. In exact rationals: the admissible stress cone of §4.3, where Lemma 4.5’s square-root-free form (4.22) agrees with the relaxed condition on a sweep of 25,200 points (7,234 admissible, 17,966 not), a planted sign flip disagrees at 10,456 of them, and Proposition 7.5’s printed amplitudes are real exactly on the reference cone at 1,640 points (224 inside); and the

swirl equation above. In floating point, deciding nothing, as the record says: the axis initial-value problem of Appendix B at representative data the paper never fixes, whose solved profile tends to the closed form of Proposition B.2 as $1/\Lambda$, with (B.17) holding from $\Lambda = 10^6$ on. The record also keeps this battery’s own first bug, a wrong derivative factor $((-h)_m$ for $(-1)^m(h)_m$) that made a first run report H failing its equation; it is now a red control, so that a scratch log reading FAIL against a Millennium claim is not mistaken for a finding.

3.8 The companion theorem

The same repository formalises Theorem 1.1 of [3]: there is $u_0 \in C_{c,\sigma}^\infty(\mathbb{R}^3)$ with $0 < T^*(u_0) < \infty$ whose smooth unforced Euler solution has $\limsup_{t \uparrow T^*} \|\nabla u(t)\|_{L^\infty} = \infty$ and $\int_0^{T^*} \|\text{curl } u(t)\|_{L^\infty} dt = \infty$, a breakdown with no force, read in full [19]. Of its two formal shapes, `exists_compact_smooth_euler_singularity` asserts a nonzero smooth compactly supported datum, $0 < T^* \leq 1$, a uniform energy bound on the whole lifespan, maximality as an equivalence, infinite $\limsup$ of the C^1 norm and infinite Beale–Kato–Majda integral at T^* , and no global smooth finite-energy solution; `euler_breakdown_R3` is the strictly weaker non-existence form, equivalent to blowup for these data only because the formal class imposes square integrability and bounded energy.

The step where such arguments die is the transfer from the approximants U_j to the limiting datum, Euler’s stability constant $\exp(C \int \|\nabla U_j\|_\infty)$ being the diverging quantity; the reading finds §6.2 complete as written, because it compares each U_j with the hypothetical smooth solution of the limiting datum under the contradiction hypothesis $T^*(u_0) > T_\infty$, so the Gronwall constant is $C \sup \|u\|_{H^4}$, fixed before j is chosen. The reading’s sharpest candidate error, (3.16) on p. 13 apparently printing $(k \cdot C_*)^{n+1}$ with $C_* = 80$, false at $n = 0$, was WITHDRAWN: the page rendered at 150 dpi shows $(k^{C_*})^{n+1}(n!)^2$; the extraction had flattened the superscript, the second such artifact of this audit. The record lists 3 checks the reading transcribed and did not run: the frozen-frame WKB model as an exact ODE system, every scale inequality of §5.5 and (6.1)–(6.2) in exact logarithmic arithmetic, and the symbolic identities of §3 including the trace check that fixes the -2 in (3.14). The Euler result is audited separately and not folded into the Navier–Stokes verdict.

4 What the record decides

The register row `navier-stokes-openai-2026`, recorded 2026-09-09, reads PARTIAL: the Lean certificate holds for Clay’s (C) and (D), and the finite-energy clause the paper’s Theorem 1.1 states is not in the formal statement [19]. Exactly:

1. *Held.* The formal statements are Clay’s (C) and (D) with nothing added to the competitor class and nothing dropped from the data, written by a third party before the proof existed; the competitor classes are inhabited (Section 3.1).
2. *Held.* All 2,486 modules build on this machine with the pinned toolchain; the kernel reports exactly `propext`, `Classical.choice`, `Quot.sound` for each of the 4 main declarations; Comparator accepts both configurations through Lean’s kernel and nanoda (Section 3.2).
3. *Held.* Every clause of Theorem 1.1 except the uniform energy bound and the vanishing of the force near $t = 0$ has a formal counterpart, and the formal uniqueness covers every competitor Clay admits (Section 3.3).

4. *Held, as readings.* No error, circularity or counterexample was found in six independent readings; 6 of the 12 asserted steps are discharged in Lean at full strength, 2 more strongly than the paper states; no classical obstruction refutes the construction; all 7 probe scripts pass with all 26 red controls firing; the Euler transfer step reads complete (Sections 3.4–3.8).
5. *Not decided.* The paper’s $\sup_{t < 1} \|u(t)\|_{L^2} < \infty$ and the force’s vanishing near $t = 0$ are paper claims with no formal counterpart; the first rests on the paper’s Lemma 10.4, re-derived by hand.
6. *Not decided.* The smallness parameter h is bounded numerically in neither artifact; the proof is asymptotic in a parameter no one has quantified, and nothing here makes it effective.
7. *Not decided.* The instantiation of Proposition B.2’s machinery for the actual profile was not traced to a line; Lemma 4.9’s collar inequality has no formal counterpart; the paper names no sentence for the swirl obstruction it evades; the 3 Euler checks are listed and unrun; Comparator’s process isolation did not run.
8. *Not decided.* The mathematics of the writeup beyond what the kernel and the readings establish. An audit is not a referee report, and the readings were not peer review.

5 What is not claimed

Nothing is claimed about the correctness of the mathematics beyond items 1–4 above: the kernel establishes that the proof terms check against the third party’s statement, the readings that nobody here found an error, and the verdict on the theorem is the build’s and the kernels’, not the reading’s. No numerical computation here certifies anything: the paper contains no numerics and none of its constants is a number. Nothing is claimed about priority: the record holds the dates each party states (Section 6) and adjudicates nothing, and the concurrent results [10] are different theorems from the one audited here. Nothing is claimed about the prize: the papers are published and not peer-reviewed, the Clay Mathematics Institute had said nothing at the time of the record, and OpenAI has said it does not intend to claim it [4]. The word *certified* is used here only for what a kernel checked.

6 The dates, as the parties state them

The record holds 11 dates, each from the document named in it: OpenAI’s announcement [4], Buckmaster’s statement and the concurrent papers [10], Tao’s posts [11], or a repository’s own timestamps [5, 6]. Nothing here weighs them [19].

2026-05-15: DeepMind’s Formal Conjectures adds the Lean statement of the four Clay alternatives (#1457); **2026-07-27:** last change to that file (#4643, divergence lemmas); the revision OpenAI pinned; **2026-08-15:** Alpöge–Buckmaster obtain smooth-forcing blowup for Boussinesq and Euler (their statement); **2026-08-22:** Alpöge–Buckmaster’s Lean verification (their statement); **2026-08-28:** OpenAI begins training the internal model used (the announcement); **2026-09-01:** OpenAI launches the agent effort after ‘a rumor that two Millennium Prize problems had been resolved’ (the announcement); **2026-09-03:** Tao’s six-part post on the opportunity cost of an autonomous solution; Buckmaster writes to OpenAI (his statement); **2026-09-05:** OpenAI’s agents ‘arrive at their resolution’, 88 hours in (the announcement); Tao’s clarification: ‘not aware of any significant developments’; **2026-09-06:** Lean formalization complete, 17 hours later (the announcement); the two calls with

Buckmaster (his statement); **2026-09-08 03:58 UTC**: Alpöge–Buckmaster publish IPM, Boussinesq and Euler with smooth forcing, Lean at `tristanbuckmaster/fluid_lean` (repository created 04:03 UTC); **2026-09-08 10:53 UTC**: `openai/NavierStokesAndEuler` created; its single commit at 10:57 UTC; the announcement the same day.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The records are in `corpus/navier-stokes/` [19]: `lean-repo.json` is computed from a clone at 8937a8f4 by `tools/pin-navierstokes-lean.js`, `build.json` by `tools/record-navierstokes-build.js`, and `probes.json` by `python3 instruments/navierstokes/battery.py`, which re-runs the 7 probe scripts with their 26 red controls. The macros of this document are written by `node tools/paper-numbers/navier-stokes-audit.js` from those records at repository commit `eece01c1497a`; the timing fields of `probes.json`, which change on every run, are never read. Rebuilding the Lean repository needs the clone and toolchain the two tools name and about 140 minutes on an eight-core machine; nothing here depends on re-running it, and every number above is read from the record of the run of 2026-09-09.

References

- [1] C. L. Fefferman, Existence and smoothness of the Navier–Stokes equation, Clay Mathematics Institute, 2000, with errata, <https://www.claymath.org/wp-content/uploads/2022/06/navierstokes.pdf>, pinned 2026-09-09.
- [2] OpenAI, Finite time blowup for Navier–Stokes, 166 pp., 8 September 2026, <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>, pinned sha256 0e779481...
- [3] OpenAI, Finite time blowup for the Euler equation, 57 pp., 8 September 2026, <https://cdn.openai.com/pdf/315b36cd-ec98-4023-8342-93345194ece1/euler.pdf>.
- [4] OpenAI, On the Navier–Stokes Millennium Prize Problem, 8 September 2026, <https://openai.com/index/navier-stokes-solution/>, pinned 2026-09-09 via the Internet Archive.
- [5] OpenAI, NavierStokesAndEuler: Lean 4 formalizations, <https://github.com/openai/NavierStokesAndEuler>, commit 8937a8f4, 2026-09-08.
- [6] Google DeepMind, Formal Conjectures, `FormalConjectures/Millennium/NavierStokes.lean`, <https://github.com/google-deepmind/formal-conjectures>, revision 8bf45ed7; the statement added 2026-05-15 in #1457.
- [7] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, in: Automated Deduction – CADE 28, LNCS 12699, Springer, 2021, 625–635; The mathlib Community, The Lean mathematical library, in: Proc. CPP 2020, ACM, 367–381; Mathlib revision 85e3a25e as pinned.
- [8] Lean FRO, Comparator, <https://github.com/leanprover/comparator>, revision 19e111e2; `lean4export`, <https://github.com/leanprover/lean4export>, revision cacf989b; `nanoda` (`ammkrn/nanoda_lib`), a Lean kernel written in Rust, built from source for the run.
- [9] D. Córdoba, L. Martínez-Zoroa, Finite time singularities of smooth solutions for the 2D incompressible porous media (IPM) equation with a smooth source, arXiv:2410.22920v3 (2025).
- [10] L. Alpöge, T. Buckmaster, Blowup for the Boussinesq equations with smooth forcing, and Blowup for the Euler equations with smooth forcing; with M. P. Coiculescu, Extending the Córdoba–Martínez-Zoroa IPM blow-up to uniformly space-time smooth forcing; T. Buckmaster, statement of 8 September 2026; all at <https://cims.nyu.edu/~tristanb/>, pinned 2026-09-09.

- [11] T. Tao, posts of 3 and 5 September 2026, <https://mathstodon.xyz/@tao/117207849921390904>, pinned 2026-09-09.
- [12] J. Leray, Sur le mouvement d'un liquide visqueux emplissant l'espace, *Acta Math.* 63 (1934), 193–248.
- [13] J. Serrin, On the interior regularity of weak solutions of the Navier–Stokes equations, *Arch. Rational Mech. Anal.* 9 (1962), 187–195.
- [14] L. Caffarelli, R. Kohn, L. Nirenberg, Partial regularity of suitable weak solutions of the Navier–Stokes equations, *Comm. Pure Appl. Math.* 35 (1982), 771–831.
- [15] J. T. Beale, T. Kato, A. Majda, Remarks on the breakdown of smooth solutions for the 3-D Euler equations, *Comm. Math. Phys.* 94 (1984), 61–66.
- [16] L. Escauriaza, G. Seregin, V. Šverák, $L_{3,\infty}$ -solutions of Navier–Stokes equations and backward uniqueness, *Russian Math. Surveys* 58 (2003), 211–250.
- [17] G. Koch, N. Nadirashvili, G. Seregin, V. Šverák, Liouville theorems for the Navier–Stokes equations and applications, *Acta Math.* 203 (2009), 83–105.
- [18] C.-C. Chen, R. M. Strain, H.-T. Yau, T.-P. Tsai, Lower bound on the blow-up rate of the axisymmetric Navier–Stokes equations, *Int. Math. Res. Not. IMRN* 2008, Art. ID rnn016.
- [19] cert-machine, the records of the audit in `corpus/navier-stokes/` (`MANIFEST.json`, `lean-repo.json`, `build.json`, `nonvacuity.json`, `fidelity-clay-vs-lean.md`, `dissection.md`, `audit.json`, `euler.md`, `probes.json`), all of 2026-09-09; the register row `navier-stokes-openai-2026` in `certs/claims-ledger.json`; and the page built from them, <https://carlostoledo.co/reports/navier-stokes.html>.